Trust-Aware Professional Identity Representation in AI-Mediated Systems

Shubham Anandani
School of Computer Science
Software and Societal Systems Department
Carnegie Mellon University
Pittsburgh, PA, USA
0009-0005-2203-2235

***Abstract*—As artificial intelligence systems increasingly participate in evaluating, ranking, and recommending professionals, digital identity representations play a critical role in shaping opportunity access and trust formation. Traditional profiles emphasize static credentials, yet fail to capture trust-related signals that humans implicitly rely upon. This paper proposes a trust-aware conceptual framework for professional identity representation, grounded in established trust theory and adapted for AI-mediated environments. Building on organizational trust models, we introduce a four-dimensional construct—Authenticity, Credibility, Empathy, and Uniqueness (ACE-U)—and analyze its implications for the design of scalable, machine-interpretable identity systems. We discuss how each dimension can be operationalized through observable signals, examine interactions and trade-offs among the dimensions, and outline system-level considerations for responsible AI evaluation.**

***Index Terms*—Digital Identity, Trust Modeling, Professional Profiles, AI Systems, Human-Centered Computing**

I. Introduction

Professional identity is increasingly expressed and interpreted through digital systems. Automated screening, ranking, and recommendation mechanisms now influence hiring, collaboration, and reputation at scale. While these systems process structured attributes such as education and employment history, trust formation remains under-specified in computational identity models.

Human evaluators rely on nuanced trust cues when assessing others, including perceived authenticity, demonstrated competence, and behavioral signals such as how individuals treat and support their peers. Translating these cues into digital identity representations suitable for AI systems presents both a challenge and an opportunity. This paper addresses the following question: *How can professional identity systems represent trust in a manner that is interpretable by both humans and AI-driven evaluators?*

Beyond individual decisions, these identity representations have system-wide implications. At scale, the way AI systems encode and prioritize identity attributes can shape labor market access, visibility of non-traditional careers, and the perceived legitimacy of emerging professional roles. Designing representations that capture trust-related signals without reinforcing bias is therefore both a technical and socio-technical challenge.

We propose a conceptual framework that synthesizes trust theory with the design requirements of large-scale identity systems. The ACE-U framework treats trust as multi-dimensional and explicitly incorporates differentiation, supporting richer and more accountable AI-mediated evaluation.

II. Background and Related Work

Trust has been extensively studied in organizational behavior, human–computer interaction (HCI), and social psychology. In organizational contexts, trust is often modeled as a willingness to be vulnerable to another party's actions, based on positive expectations of their intentions and behavior [4]. Prior work distinguishes between cognitive trust (beliefs about another's competence and reliability) and affective trust (perceptions of care, concern, and relational warmth). These dimensions jointly influence cooperation, delegation, and credibility assessment in professional settings.

In HCI and automation research, trust is treated as a determinant of appropriate reliance on technology. Lee and See [2] highlight how transparency, predictability, and feedback shape human trust in automated systems and emphasize the risks of over- or under-reliance when trust is miscalibrated. Analogously, professionals and organizations now rely on AI-mediated identity systems—ranking models, recommender systems, and matching algorithms—whose internal criteria for "trustworthy" or "high-potential" profiles are often opaque.

A. Digital Identity and Reputation Systems

Digital identity and reputation systems introduce additional complexities. Work on online marketplaces and reputation platforms shows how structured feedback (ratings, reviews, endorsements) can serve as partial trust proxies [5], [6], but also how they can entrench bias, be gamed, or fail to capture important qualitative aspects of contribution. Research on open-source development platforms demonstrates that professionals assess credibility through multi-faceted signals beyond simple metrics, including contribution patterns, peer recognition, and project context [7]. Professional identity systems typically foreground verifiable credentials and employment history while underrepresenting behavioral, contextual, and community-oriented signals.

As AI systems increasingly act as intermediaries, the absence of explicit, multi-dimensional trust modeling in identity schemas risks brittle or biased outcomes: small perturbations

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in credentials can dominate nuanced, longitudinal signals of reliability, care, or differentiation. Many computational approaches implicitly reduce trust to a scalar score—for example, a reputation rating, risk score, or predicted performance metric. While convenient for ranking, such reductions obscure important trade-offs among different bases of trust. An individual may be highly credible but perceived as inauthentic; another may be deeply empathetic and authentic but lack clear credentials in a given domain. Collapsing these distinctions into a single dimension hinders interpretability and can degrade decision quality when systems are deployed in diverse, nuanced professional contexts [8].

B. Algorithmic Hiring and Bias

Research on algorithmic hiring and socio-technical systems has documented how AI-mediated screening and ranking can amplify historical inequalities through biased training data, proxy variables, and feedback loops [9], [10]. These systems often perpetuate existing power structures by privileging easily quantifiable signals (such as institutional prestige) over harder-to-measure qualities (such as mentorship or community contribution). The ACE-U framework targets an earlier layer in this stack: the design of identity schemas and signals that ranking and matching systems consume. By making trust dimensions explicit and multi-faceted, ACE-U aims to provide designers with conceptual tools to recognize and mitigate reductionist representations.

C. Fairness and Interpretability

Recent work in fairness and interpretability offers metrics and methods for evaluating and explaining machine learning models [11], [12]. However, these approaches typically assume an existing feature representation. ACE-U can be seen as a domain-specific proposal for what features a trust-aware professional identity representation should expose to downstream fairness and interpretability tools. By structuring identity along authenticity, credibility, empathy, and uniqueness dimensions, systems can support more transparent audits and stakeholder-intelligible explanations.

Frances Frei's Trust Triangle [1] offers a compact yet expressive alternative: trust tends to erode when any one of authenticity, credibility, or empathy is absent. This decomposition provides a conceptual bridge between qualitative trust theory and structured, machine-interpretable representations. However, in large-scale professional ecosystems, many individuals may satisfy basic thresholds on these dimensions. Identity systems must therefore support not only "can I trust this person?" but also "how is this person meaningfully different from others I could trust?", motivating the explicit inclusion of Uniqueness.

III. Trust as a Multi-Dimensional Construct

Frei's Trust Triangle identifies three core dimensions of trust: Authenticity, Credibility (termed Logic in the original model), and Empathy [1]. Authenticity reflects perceived genuineness, credibility captures competence and reliability, and empathy represents care and understanding toward others. This model has been widely adopted in leadership and organizational contexts.

TABLE I
ACE-U trust dimensions for professional identity

Dimension	Description
Authenticity	Alignment between expressed identity and observable actions or outputs
Credibility	Demonstrated competence through verifiable experience and achievements
Empathy	Signals of social awareness, collaboration, and community engagement
Uniqueness	Distinctive skills, perspectives, or career paths that enable differentiation

While these dimensions remain applicable, digital professional identity systems introduce an additional pressure: differentiation. At scale, trust alone is insufficient without distinctiveness. AI systems must differentiate among many credible and authentic individuals in order to surface diverse and complementary matches for roles, teams, and projects.

We therefore extend the traditional trust model with a fourth dimension: Uniqueness. Uniqueness captures distinctive skills, perspectives, or trajectories that make a professional meaningfully different in a given context. Importantly, uniqueness in ACE-U is bounded to domain-relevant differentiation—such as uncommon but useful skill combinations or cross-disciplinary experience—rather than novelty for its own sake, which could bias systems toward eccentric but unqualified profiles. In combination, Authenticity, Credibility, Empathy, and Uniqueness (ACE-U) provide a structured lens for thinking about professional identity in AI-mediated environments.

IV. The ACE-U Framework

The proposed ACE-U framework consists of four dimensions:

The ACE-U framework treats professional identity as a point in a four-dimensional space, where each dimension can be estimated from observable signals with varying confidence. The intent is not to reduce individuals to a fixed vector, but to provide a scaffolding for designing identity schemas, features, and system behaviors. Importantly, these dimensions interact: attempts to maximize one may inadvertently suppress another. For example, highly optimized, keyword-dense profiles may increase perceived credibility to an AI system while decreasing perceived authenticity for human reviewers.

Authenticity, in this framing, is less about personal style and more about coherence over time. A professional whose self-descriptions, work artifacts, and third-party feedback tell a consistent story is more likely to be perceived as authentic than one whose claimed skills fluctuate rapidly or diverge from observable outputs. Credibility builds on this by layering externally verifiable evidence—degrees, certifications, track records—on top of that coherent narrative.

Empathy and Uniqueness add social and differentiation dimensions. Empathy connects identity to community: does

this professional engage with others, recognize contributions, and participate in shared problem-solving? Uniqueness captures the extent to which an identity is distinguishable and potentially complementary in a given ecosystem—through uncommon skill combinations, non-linear trajectories, or novel perspectives. For AI-mediated systems, explicitly modeling these dimensions allows rankings and recommendations to move beyond "more of the same" and support diverse, trustworthy matches.

Conceptually, ACE-U can be visualized as a tetrahedral space in which different professional archetypes occupy different regions: high-credibility, low-uniqueness experts; high-authenticity, high-empathy community builders; high-uniqueness innovators with emerging credibility; and so on. This mental model can guide designers in deciding which regions their systems implicitly favor.

V. Operationalizing Trust in Identity Systems

Each ACE-U dimension can be represented through observable signals. This section outlines example signals and design considerations.

A. Authenticity

Authenticity-related signals may include longitudinal consistency of skills and roles, alignment between stated interests and actual projects, and the presence of authored artifacts (such as talks, code, or writing) that substantiate claims. Sudden profile shifts that are not supported by new work artifacts may be treated as lower-confidence transitions, prompting systems to down-weight or flag them for human review rather than silently incorporating them as ground truth.

From a systems perspective, designers must decide how aggressively to normalize or suggest edits to profiles. Overly prescriptive templates can homogenize language and inadvertently obscure authentic voice. A trust-aware approach would preserve space for narrative elements and user-controlled descriptions while still extracting structured features for AI consumption.

B. Credibility

Credibility is the dimension most aligned with existing professional schemas. Degrees, titles, organizational affiliations, and tenure are readily structured and already heavily used in ranking and matching. The ACE-U framing suggests two refinements. First, credibility should be contextual: the same credential may be highly predictive in one domain and only weakly informative in another. Second, systems should distinguish between directly verified signals (e.g., institution-issued, cryptographically signed credentials) and indirectly inferred ones (e.g., self-reported skills endorsed by peers), exposing this distinction to downstream consumers.

AI-mediated systems can also incorporate uncertainty estimates for credibility. Rather than treating all profiles as equally well-understood, models could indicate when a profile's credibility is under-determined (for example, early-career individuals or career-changers) and adjust ranking logic accordingly, for instance by leaning more on authenticity or empathy signals.

C. Empathy

Empathy is less about sentiment and more about orientation toward others in the professional ecosystem. Observable signals include patterns of interaction such as congratulating peers on milestones, offering constructive feedback, mentoring others, and amplifying under-recognized work. These behaviors often show up in comments, endorsements with substantive justifications, participation in community forums, or contributions to collective resources.

Naïve metrics—such as raw counts of connections or endorsements—are poor proxies for empathy. A trust-aware system would focus on the quality, reciprocity, and diversity of interactions. For example, a small number of detailed, domain-relevant endorsements from collaborators may be more informative than a large number of generic acknowledgments. Designing features that capture these distinctions, while respecting privacy and avoiding intrusive surveillance, is an open research challenge but central to modeling empathy responsibly.

D. Uniqueness

Uniqueness can be operationalized through measures of rarity and complementarity. Rarity-based signals may identify combinations of skills, experiences, or domains that occur infrequently in the overall population. Complementarity-based signals consider how an individual's profile relates to a target context, such as a team or project: a candidate whose background fills gaps or brings new perspectives may be uniquely valuable even if conventional credentials are modest.

However, emphasizing uniqueness without a trust-aware foundation can lead to unintended consequences, such as overrewarding novelty for its own sake. The ACE-U framework positions uniqueness as valuable when anchored in sufficient authenticity, credibility, and empathy. For AI systems, this implies multi-objective ranking and matching strategies that balance these dimensions rather than optimizing a single novelty score.

E. Data and Governance Considerations

Operationalizing ACE-U requires data that may be sensitive, contextual, or unevenly distributed. Systems must address consent, visibility controls, and mechanisms for individuals to contest or correct inferred signals. Governance processes—such as regular audits of feature distributions across demographic groups—are necessary to ensure that trust-related features do not amplify existing inequities.

VI. Implementation and Evaluation Pathway

While this work is conceptual, the ACE-U framework admits a straightforward pathway toward implementation and validation. This section outlines how such a system could be constructed and empirically evaluated.

A. Reference Implementation

A minimal ACE-U system would comprise four layers:

1) **Data ingestion:** Collect profile text, employment and education history, endorsements, interaction logs (comments, reactions), and network features (connections, clustering, diversity of ties).
2) **Feature extraction and modeling:** Extract signals for each dimension. Authenticity can be estimated from longitudinal consistency of claims and overlap between stated skills and observable artifacts. Credibility aggregates institutional affiliations, verified credentials, tenure, and external recognitions. Empathy captures frequency and quality of other-directed actions (peer congratulations, feedback, mentorship mentions). Uniqueness measures rarity of skill combinations and cross-domain links.
3) **ACE-U estimation:** A supervised or weakly-supervised model maps extracted features to ACE-U scores, trained with expert annotations or proxy labels derived from historical outcomes (e.g., hiring success, collaboration satisfaction).
4) **Application endpoints:** Expose ACE-U vectors and explanations to downstream services including ranking, recommendation, auditing dashboards, and user-facing feedback.

This modular architecture allows ACE-U estimation to evolve independently of specific applications while providing a single, inspectable representation of trust-related identity dimensions.

B. Evaluation Strategy

Validating ACE-U requires both offline and field studies:

- **Offline study:** Sample several thousand profiles; have human raters score authenticity, credibility, empathy, and uniqueness; compare model estimates to human judgments using correlation and ranking metrics.
- **A/B or simulation study:** Integrate ACE-U into a toy ranking scenario (e.g., candidate recommendation) and compare a credibility-only baseline to an ACE-U-aware ranking on diversity of surfaced profiles, perceived fairness, and decision quality.
- **Qualitative validation:** Conduct interviews with professionals and hiring managers to assess whether ACE-U explanations ("you are high in credibility, moderate in uniqueness") match self-perception and support actionable feedback.

Such an evaluation would establish the empirical grounding needed to transition ACE-U from conceptual framework to deployable system.

VII. Implications for AI-Mediated Evaluation

Beyond immediate improvements to ranking or matching accuracy, incorporating ACE-U into identity representations can reshape how AI systems are evaluated and deployed. Evaluation metrics can move from purely outcome-based measures (such as click-through or short-term hiring rates) toward metrics that reflect alignment with trust-related goals, such as long-term retention, team cohesion, or diversity of hired profiles.

A trust-aware evaluation pipeline might:

1) Encode profiles into ACE-U representations with associated uncertainty measures.
2) Use multi-objective optimization to generate candidate rankings that trade off between relevance, trust alignment, and diversity along the uniqueness dimension.
3) Present structured explanations to human decision-makers that highlight which ACE-U dimensions primarily drove a given recommendation.
4) Log outcomes and feedback to iteratively refine both feature extraction and ranking logic.

This approach encourages appropriate reliance: humans can see when an AI system is basing its recommendation primarily on credibility, for example, and decide whether that is adequate for the decision at hand, or whether authenticity, empathy, or uniqueness should carry more weight.

A. Risks and Failure Modes

Trust-aware features can themselves be gamed or misused. Over-emphasis on authenticity signals, for instance, could punish individuals who must compartmentalize aspects of their identity for safety or cultural reasons. Similarly, community engagement metrics might disadvantage professionals with limited bandwidth for online participation, such as caregivers. Systems must therefore be designed with sensitivity to structural constraints and support configurable weighting schemes that reflect different organizational values and contexts.

VIII. Limitations and Future Work

This paper presents a conceptual framework rather than an empirical validation. Future empirical work could take several forms. Controlled experiments might compare AI-mediated selection processes with and without ACE-U-informed representations, measuring effects on decision quality, perceived fairness, and user trust. Longitudinal field deployments could examine how exposing ACE-U-based explanations influences both professionals' self-presentation strategies and organizations' recruitment behaviors.

Another avenue is participatory design: involving professionals, hiring managers, and affected communities in shaping which signals should count as authenticity, empathy, or uniqueness in their contexts. Such work could surface domain-specific adaptations of ACE-U (for example, for academic hiring versus open-source collaboration) and refine the framework beyond the generic treatment offered here.

IX. Conclusion

As AI systems increasingly mediate professional evaluation, trust-aware identity representation becomes a foundational requirement. By extending established trust theory with a

differentiation-focused dimension, the ACE-U framework provides a structured approach for designing identity systems that are both human-aligned and machine-interpretable.

Explicitly modeling Authenticity, Credibility, Empathy, and Uniqueness encourages designers to move beyond single-score representations and to consider how different bases of trust interact in practice. Our aim is to inform future research and system design at the intersection of trust, identity, and AI, and to support professional ecosystems in which digital representations more faithfully capture the qualities humans rely on when deciding whom to trust and with whom to work.

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